

Question Number	Answer				Additional guidance	Mark
6(a)	Structure	Animal cell ✓ or X	Plant cell ✓ or X	Prokaryotic cell ✓ or X	1 mark per correct row do not credit hybrid tick/cross	(4)
	amyloplast	X	✓	X		
	circular DNA	✓	✓	✓		
	mitochondria	✓	✓	X		
	nucleolus	✓	✓	X		

Question Number	Answer	Mark
6(b)	The only correct answer is A $4.2 \times 10^{-3} \mu\text{m}^3$ <i>B is not correct because that is to an incorrect power of 10</i> <i>C is not correct because that is using the diameter</i> <i>D is not correct because that is using the diameter</i>	(1)

Question Number	Answer	Mark
6(c)(i)	The only correct answer is C 2 only <i>A is not correct because sperm cells contain half the DNA and starch does not release energy for sperm cell motility</i> <i>B is not correct because starch does not release energy for sperm cell motility</i> <i>D is not correct because starch does not release energy for sperm cell motility</i>	(1)

Question Number	Answer
*6(c)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • fertilised egg cells have a thinner zona pellucida (ZP) than unfertilised egg cells / converse • correct description of one piece of data from table • thicker the ZP then the lower the {probability / chance} of fertilisation / thicker ZP increases chance of infertility • relevant comment regarding {how egg cells are fertilised / sperm cell adaptation} • SD overlap so {there is not a significant difference / comparative conclusions are invalid} / relevant comment about size of SD linked to {reliability/validity} • some sperm cells may contain less {acrosin / enzyme for acrosome reaction} • problems digesting the zona pellucida with a lower {acrosin / enzyme} concentration • it takes longer for the enzyme to digest a thicker ZP / may not be sufficient enzyme to digest a thicker ZP / enzymes might not be able to digest all ZP • low sperm count could reduce the {probability / chance} of fertilisation • fewer egg cells are fertilised if there is a longer time period between ejaculation and fusing with egg cell membrane • a long time period could result in sperm {losing motility / not having enough energy / dying} • increase (probability / chance) of fertilisation using IVF (compared to non-IVF fertilisation) / injecting sperm would increase number of fertilised egg (cells) • sperm cells do not need to digest ZP if they are injected into egg cell / helps sperm cells {penetrate membrane / reach nucleus} • IVF technique can inject {non-motile / deformed / low acrosin} sperm cells into egg cell • consideration of {ethical issues / expense} from using IVF • consideration of possible increase in cancer risk from IVF hormones / risk of polyspermy / relevant comments regarding health risk

	(6)		
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Limited scientific judgement made with a few strengths/weaknesses identified. A conclusion may be attempted, demonstrating isolated elements of biological knowledge and understanding but with limited evidence to support the conclusion.	Limited conclusion covering at least one of the following: - the effect on mean ZP thickness from the table - the fertilisation process linked to thickness of {ZP / timescale} - effects of injecting sperm cells
Level 2	3-4	A scientific judgement is made through the application of relevant evidence, with strengths and weaknesses identified. A conclusion is made, demonstrating linkages to elements of biological knowledge and understanding, with occasional evidence to support the judgement being made.	Conclusion covering at least two of the following: - the effect on mean ZP thickness from the table - the fertilisation process linked to thickness of {ZP / timescale} - effects of injecting sperm cells
Level 3	5-6	A scientific judgement is made which is supported throughout by sustained application of relevant evidence from the analysis and interpretation of the scientific information. A conclusion is made, demonstrating sustained linkages to biological knowledge and understanding with evidence to support the judgement being made.	Conclusion covering at least three of the following: - the effect on mean ZP thickness from the table - the fertilisation process linked to thickness of {ZP / timescale} - effects of injecting sperm cells